

Answer

Q1: $\pi_{sid, sname, age}(\sigma_{rating < 10}(S_2))$

第一步: $\pi_{sid, sname, age}(\sigma_{rating < 10}(S_2))$

“在 S2 表中选择，条件：排名小于 10”

sid	sname	rating	age
28	yuppy	9	35
31	lubber	8	55.5
44	Guppy	5	35
<u>58</u>	<u>rusty</u>	<u>10</u>	<u>35</u>

第二步: $\pi_{sid, sname, age}(\sigma_{rating < 10}(S_2))$

“投影 保留字段：sid, sname, age”

sid	sname	rating	age
28	yuppy	<u>9</u>	35
31	lubber	<u>8</u>	55.5
44	Guppy	<u>5</u>	35

最终答案:

sid	sname	age
28	yuppy	35
31	lubber	55.5
44	Guppy	35

Q2: $\pi_{sname, rating}(\sigma_{S_1.sid > R_1.sid}(S_1 \times R_1))$

第一步: $\pi_{sname, rating}(\sigma_{S_1.sid > R_1.sid}(S_1 \times R_1))$

2 个操作 笛卡尔乘积+选择 (条件: S1.sid>R1.sid)

$S_1.sid$	sname	rating	age	$R_1.sid$	bid	day
<u>22</u>	<u>dustin</u>	<u>7</u>	<u>45</u>	<u>22</u>	<u>101</u>	<u>10/10/96</u>
<u>22</u>	<u>dustin</u>	<u>7</u>	<u>45</u>	<u>58</u>	<u>103</u>	<u>11/12/96</u>
31	lubber	8	55.5	22	101	10/10/96
<u>31</u>	<u>lubber</u>	<u>8</u>	<u>55.5</u>	<u>58</u>	<u>103</u>	<u>11/12/96</u>
58	rusty	10	35.0	22	101	10/10/96
<u>58</u>	<u>rusty</u>	<u>10</u>	<u>35.0</u>	<u>58</u>	<u>103</u>	<u>11/12/96</u>

第二步: $\pi_{sname, rating}(\sigma_{S_1.sid > R_1.sid}(S_1 \times R_1))$

投影, 保留字段 sname, rating:

<u>$S_1.sid$</u>	sname	rating	<u>age</u>	<u>$R_1.sid$</u>	<u>bid</u>	<u>day</u>
<u>31</u>	lubber	8	<u>55.5</u>	<u>22</u>	<u>101</u>	<u>10/10/96</u>
<u>58</u>	rusty	10	<u>35.0</u>	<u>22</u>	<u>101</u>	<u>10/10/96</u>

最终答案:

sname	rating
lubber	8
rusty	10

Q3: $\pi_{sid, bname}(\sigma_{bid}(R_1 \times B_1))$

第一步: $\pi_{sid, bname}(\sigma_{bid}(R_1 \times B_1))$

$\sigma_{bid}(R_1 \times B_1) = R_1 \bowtie_{bid} B_1 \neq \sigma_{R_1.bid=B_1.bid}(R_1 \times B_1)$:

两个表格笛卡尔乘积后按条件选择（等价于带条件的连接操作）

等值连接，条件：两个表格 bid 相等，但结果中的 bid 字段保留一个

sid	bid	day	bname	color
22	101	10/10/96	tiger	red
58	103	11/12/96	lion	green

第二步: $\pi_{sid, bname}(\sigma_{bid}(R_1 \times B_1))$

投影，保留字段 sid, bname:

sid	<u>bid</u>	<u>day</u>	bname	<u>color</u>
22	<u>101</u>	<u>10/10/96</u>	tiger	<u>red</u>
58	<u>103</u>	<u>11/12/96</u>	lion	<u>green</u>

最终答案:

sid	bname
22	tiger
58	lion

Q4: $S_2 / (\pi_{sid,age}(S_2) / \pi_{sid}(\pi_{bid=103}(R_1)))$

第一步: $S_2 / (\pi_{sid,age}(S_2) / \pi_{sid}(\sigma_{bid=103}(R_1)))$

先分别投影,然后做除法

$\pi_{sid,age}(S_2)$ $\pi_{sid}(\sigma_{bid=103}(R_1))$ 结果

sid	age	除以	sid	等于	age
28	35.0		58		35
31	55.0				
44	35.0				
58	35.0				

第二步: $S_2 / (\pi_{sid,age}(S_2) / \pi_{sid}(\pi_{bid=103}(R_1)))$

做除法

最终答案:

sid	sname	rating	age	除以	age	等于
28	yuppy	9	35.0		35	
31	lubber	8	55.0			
44	guppy	5	35.0			
58	rusty	10	35.0			

sid	sname	rating
28	yuppy	9
44	guppy	5
58	rusty	10

Q5: $(S_2 \bowtie^* R_1) \underline{\cup} (B_1 \bowtie^* R_1)$

第一步: $(S_2 \bowtie^* R_1) \underline{\cup} (B_1 \bowtie^* R_1)$:

基于 sid 右外连接, 只保留 R_1 中所有的元组, 无法匹配的内容 null

sid	sname	rating	age	bid	day
22	null	null	null	101	10/10/96
58	rusty	10	35	103	11/12/96

第二步: $(S_2 \bowtie^* R_1) \underline{\cup} (B_1 \bowtie^* R_1)$:

基于 bid 全连接, 保留 R_1 和 B_1 中所有的元组, 无法匹配的内容 null

sid	bid	day	bname	color
22	101	10/10/96	tiger	red
58	103	11/12/96	lion	green
null	105	null	hero	blue

第三步: $(S_2 \bowtie^* R_1) \underline{\cup} (B_1 \bowtie^* R_1)$:

外并: 字段是两个表格的字段并集, 不存在值填 full

sid	sname	rating	age	bid	day	bname	color
22	null	null	null	101	10/10/96	null	null
58	rusty	10	35	103	11/12/96	null	null
22	null	null	null	101	10/10/96	tiger	red
58	null	null	null	103	11/12/96	lion	green
null	null	null	null	105	null	hero	blue

Q6: $\pi_{sname, rating, age, bname, color} \left(((S_1 \cup S_2) \bowtie_{sid} R_1) \bowtie_{bid} \sigma_{color=red}(B_1) \right)$

第一步:

$\pi_{sname, rating, age, bname, color} \left(((\textcolor{red}{S}_1 \cup \textcolor{red}{S}_2) \bowtie_{sid} R_1) \bowtie_{bid} \sigma_{color=red}(B_1) \right)$

集合并集，重复项去除

sid	sname	rating	age
22	dustin	7	45
31	lubber	8	55.5
58	rusty	10	35
28	yuppy	9	35
44	guppy	5	35
58	rusty	10	35

第二步:

$\pi_{sname, rating, age, bname, color} \left(((S_1 \cup S_2) \bowtie_{\textcolor{red}{sid}} \textcolor{red}{R}_1) \bowtie_{bid} \sigma_{color=red}(B_1) \right)$

等值连接，条件: sid

sid	sname	rating	age	bid	day
22	dustin	7	45	101	10/10/96
58	rusty	10	35	103	11/12/96

第三步:

$$\pi_{sname, rating, age, bname, color} \left(\left((S_1 \cup S_2) \bowtie_{sid} R_1 \right) \bowtie_{bid} \sigma_{color=red}(B_1) \right)$$

选择, 条件 color=red

bid	bname	color
101	tiger	red

第四步:

$$\pi_{sname, rating, age, bname, color} \left(\left((S_1 \cup S_2) \bowtie_{sid} R_1 \right) \bowtie_{bid} \sigma_{color=red}(B_1) \right)$$

等值连接 条件: bid

sid	sname	rating	age	bid	day	bname	color
22	dustin	7	45	101	10/10/96	tiger	red

第五步:

投影 保留字段: 5 个

sid	sname	rating	age	bid	day	bname	color
22	dustin	7	45	101	10/10/96	tiger	red

最终答案

sname	rating	age	bname	color
dustin	7	45	tiger	red

Q7: $\pi_{sid}(S_2) - \pi_{sid}(\pi_{sid}(\pi_{sid,sname}(S_2)) \times \pi_{sname}(\sigma_{rating>7}(S_1)) - \pi_{sid,sname}(S_2))$

第一步：

$$\pi_{sid}(S_2) - \pi_{sid}(\pi_{sid}(\pi_{sid,sname}(S_2)) \times \pi_{sname}(\sigma_{rating>7}(S_1)) - \pi_{sid,sname}(S_2))$$

$$\pi_{sid}(\pi_{sid,sname}(S_2)) \qquad \pi_{sname}(\sigma_{rating>7}(S_1))$$

结果：

sid
28
31
44
58

×

sname
Lubber
rusty

等于

sid	sname
28	lubber
28	rusty
31	lubber
31	rusty
44	lubber
44	rusty
58	lubber
58	rusty

第二步:

$$\pi_{sid}(S_2) - \pi_{sid}(\pi_{sid}(\pi_{sid,sname}(S_2)) \times \pi_{sname}(\sigma_{rating>7}(S_1)) - \pi_{sid,sname}(S_2))$$

sid	sname		sid	sname		sid	sname
28	lubber		28	yuppy		28	lubber
28	rusty	减	31	lubber	去	28	rusty
31	lubber		44	guppy		31	rusty
31	rusty		58	rusty	等于	44	lubber
44	lubber					44	rusty
44	rusty					58	lubber
58	lubber						
58	rusty						

第三步:

$$\pi_{sid}(S_2) - \pi_{sid}(\pi_{sid}(\pi_{sid,sname}(S_2)) \times \pi_{sname}(\sigma_{rating>7}(S_1)) - \pi_{sid,sname}(S_2))$$

sid	减去	sid	=	NULL
28		28		
31		31		
44		44		
58		58		

Q8: Conver the query of Q1 to DRC formula and TRC formula.

Q1: $\pi_{sid, sname, age}(\sigma_{rating < 10}(S_2))$

域关系演算 DRC

$\{ \langle sid, sname, age \rangle \mid \langle sid, sname, age \rangle \in S_2 \wedge rating < 10 \}$

元组关系演算 TRC

$\{ t[sid, sname, age] \mid t \in S_2 \wedge t[rating] < 10 \}$

Q9: Conver the query of Q2 to DRC formula and TRC formula.

Q2: $\pi_{sname, rating}(\sigma_{S_1.sid > R_1.sid}(S_1 \times R_1))$

域关系演算 DRC

$$\{ \langle sname, rating \rangle \mid \exists S_1.sid, sname, rating, age, R_1.sid, bid, day \\ (\langle S_1.sid, sname, rating, age \rangle \in S_1 \wedge \langle R_1.sid, bid, day \rangle \in R_1 \wedge (S_1.sid > R_1.sid)) \}$$

元组关系演算 TRC

$$\{ t[sname, rating] \mid t = \langle t^s, t^r \rangle \wedge t^s \in S_1 \wedge t^r \in R_1 \wedge (t[S_1.sid] > t[R_1.sid]) \}$$

(笛卡尔积) 另一种写法供参考

$$R(A) \times S(B) = \{ t \mid \exists (u \in R) \exists (s \in S) (t[A] = u[A] \wedge t[B] = s[B]) \}$$

(笛卡尔积) 另二种写法供参考

$$R3 = \{ t \mid (\exists u) (S(t) \wedge R(u) \wedge t[3] < u[2]) \}$$

Q10: Conver the query of Q3 to DRC formula and TRC formula.

Q3: $\pi_{sid, bname}(\sigma_{bid}(R_1 \times B_1))$

域关系演算 DRC

$\{ \langle sid, bname \rangle \mid \exists sid, R_1. bid, day, B_1. bid, bname, color (\langle sid, R_1. bid, day \rangle \in R_1 \wedge \langle B_1. bid, bname, color \rangle \in B_1 \wedge (R_1. bid = B_1. bid)) \}$

元组关系演算 TRC

$\{ t[sid, bname] \mid t = \langle t^r, t^b \rangle \wedge t^b \in B_1 \wedge t^r \in R_1 \wedge t[R_1. bid] = t[B_1. bid] \}$

Q11: What' s data model? What' s data schema?

Data model is a collection of concepts and definitions for describing data.

(数据模型是用于描述数据的概念和定义的集合，比如字段)

Data schema is a description of a particular collection of data, using a given data model.

(数据模式是使用给定数据模型对特定数据集合的描述)

姓名	学号	性别	出生年份	籍贯	系别	入学时间
----	----	----	------	----	----	------

(a)

王彤	0709135	女	1989	江苏	计算机	2007
----	---------	---	------	----	-----	------

(b)

图 1-3 数据模式及其实例

Q12: Students(sid, name, , ID number cid)

Courses(cid, cname)

List all super keys, candidate keys, primary keys, foreign keys of table Students.

(1) super keys(超键)

{sid}, {ID number},

{sid, name}, {sid, ID number}, {sid, cid}, {ID number, name}, {ID number,

cid}, {sid, name, ID number}, {sid, name, cid}, {sid, ID number, cid},

{name, ID number, cid}

{sid, name, ID number, cid}

(2) candidate keys(候选键:)

{sid}, {ID number} 候选键 学号 身份证号都可以唯一标识学生

(3) primary keys(主键:只有一个)

{sid}

(4) foreign keys:(外键)

{cid}

Q13: What are the relational operations on the relational data model?

What are their differences and similarities?

Relational calculus vs Relational algebra

(1) Differences:

Relational Algebra needs to specify the order of operations; while relational calculus only needs to indicate the logic condition the result must be fulfilled.

(2) Similarities:

They are equivalent in terms of expression. SQL language can express any query that is expressible in relational algebra /calculus.